

4 Conservation

When we have nonorthogonal basis

$\hat{x}$ and $\hat{l}$, and are finding kinetic energy we replace v^2 with

$$\begin{aligned}\vec{v}_m \cdot \vec{v}_m &= \\ &= (\dot{x}\hat{x} + \dot{l}\hat{l}) \cdot (\dot{x}\hat{x} + \dot{l}\hat{l}) \\ &= \dot{x}^2 + \dot{l}^2 + 2\dot{x}\dot{l}\hat{x} \cdot \hat{l}\end{aligned}$$

a variable is cyclic if it doesn't show up in the Lagrangian.

if x is cyclic,

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}$$

is conserved, since

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial \mathcal{L}}{\partial x} = 0$$

Noethers theorem:

For every continuous symmetry of the action, there exists a corresponding conserved quantity.

$$= \int_{t_i}^{t_f} \mathcal{L} dt$$

For instance, imagine a ball falling in uniform gravity. If I move it right or left and let go, it'll fall in the same way (gravity is up and down, nothing depends on x) - so there is translational symmetry.

Applying Noethers Theorem

1. Identify a continuous symmetry of the system
2. choose a coordinate q_i that parametrizes that symmetry
3. the generalized momentum $p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$

What is conserved if

$$\frac{\partial \mathcal{L}}{\partial t} = 0?$$

We can write out

$$\frac{d}{dt} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial q} \frac{dq}{dt} + \frac{\partial \mathcal{L}}{\partial \dot{q}} \frac{d\dot{q}}{dt} + \cancel{\frac{\partial \mathcal{L}}{\partial t}}^0$$

We can use the definition of the legrangian to simplify this

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} + \frac{\partial \mathcal{L}}{\partial q} \frac{d}{dt} \dot{q}$$

This looks like the product rule

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} \right)$$

Which leads us to

$$\frac{d}{dt} \left(\left[\frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} - \mathcal{L} \right] \right) = 0$$

This inner term is the Hamiltonian!

In N dimensions, if $\mathcal{L}(q_1, q_2, \dots, q_n, \dot{q}_1, \dot{q}_2, \dots, \dot{q}_n)$

$$H = \sum_i p_i \dot{q}_i$$

Time Symmetry

There must be a continuous time symmetry for something to be time invariant, or where time changing won't change the dynamics of your free variables.